



Gunateja Pothula

Data Analyst/ Data Scientist

✉ gunateja070@gmail.com ☎ +49 15510495944 📍 Berlin, Germany
🌐 [linkedin.com/in/pothula-gunateja](https://www.linkedin.com/in/pothula-gunateja) 🐙 <https://github.com/Gunateja020>

Recent Master's graduate in Data Science with expertise in machine learning and time-series forecasting. Developed XGBoost and LSTM models on 140K+ aviation records for flight delay prediction, achieving projected cost savings of \$140K+. Proficient in Python, SQL, database design, and end-to-end model development. Seeking Data Scientist or Machine Learning Engineer roles.

Education

Masters in Data Science, Arden University

11/2024 – 11/2025

Thesis: Flight Delay Prediction & Optimization Using ML and Time-Series Analysis

Berlin, Germany

- Developed and evaluated **XGBoost and LSTM time-series models** on 140K+ flight records to predict departure delays, applying concepts from **Machine Learning** and **Artificial Intelligence and Neural Networks**.
- Performed extensive **feature engineering, hyperparameter tuning, and cross-validation** to optimize predictive performance and model reliability.
- Processed and managed large aviation datasets using **Python (Pandas, NumPy)** and scalable data-processing approaches aligned with **Big Data and Cloud Computing** principles.
- Applied statistical analysis and evaluation metrics (**RMSE, MAE, Accuracy**) using techniques from **Mathematics for Data Science** to compare model effectiveness.
- Conducted **exploratory data analysis (EDA)** to identify patterns related to weather conditions, airport congestion, and operational factors influencing delays.
- Created analytical dashboards and visualizations using **Tableau and Python libraries (Matplotlib, Seaborn)**, applying concepts from **Data Visualisation**.
- Built end-to-end data processing and modeling workflows using **Python**, demonstrating skills developed in **Programming for Data Analytics**, and produced forecasting insights supporting data-driven scheduling decisions with projected cost savings exceeding **\$140K**.

Bachelors in Communications, Visvesvaraya Technological University

06/2019 – 08/2023

Academic Project:

India

- Designed and implemented a real-time face recognition system using computer vision techniques, achieving 95%+ accuracy.
- Automated secure access for 100+ users, eliminating manual identity verification.

Professional Experience

Programmer Analyst Trainee, Cognizant Technology Solutions

02/2024 – 07/2024

India

- Designed and implemented MongoDB database schemas for healthcare patient management system handling thousands of records
- Developed and optimized database queries for efficient data retrieval, filtering, and aggregation across patient datasets
- Implemented data validation logic and quality checks ensuring 99.9% data integrity across the application
- Collaborated with cross-functional teams to translate business requirements into scalable database solutions

Machine Learning Engineer, <i>Freelance</i>	05/2023 – 01/2024	India
• Developed and deployed 5+ end-to-end machine learning solutions for sports analytics • Engineered deep learning models (Random Forest, XGBoost, CNN) for Formula 1 race prediction achieving F1-score ≥ 0.8 and RMSE < 10 • Designed interactive dashboards using Streamlit, Plotly Dash, and Tableau for real-time data visualization and business intelligence • Implemented automated data pipelines processing 100K+ records daily with 99.5% uptime • Applied unsupervised learning (K-Means, GMM) for customer segmentation and pattern recognition		

Internship Trainee, <i>Pentagon Space</i>	02/2023 – 03/2023	India
• Developed a Java-based full-stack banking application managing 100+ customer accounts and transactions • Designed SQL schemas and implemented validation logic to ensure data accuracy		

Data Science Research Assistant, <i>Visvesvaraya Technological University</i>	06/2022 – 01/2023	India
• Conducted research on deep learning applications for predictive analytics and time series forecasting • Developed Python-based data analysis scripts for processing academic research datasets with 10K+ samples • Created data visualizations and statistical reports • Collaborated with faculty on machine learning projects involving regression, classification, and clustering algorithms • Presented research findings at departmental seminars and contributed to technical documentation		

Skills

Programming & ML — C, C++, Python (pandas, NumPy, scikit-learn, TensorFlow, Keras), Java
Machine Learning & Data Analysis — XGBoost, LSTM, Time-Series Forecasting, Feature Engineering, Regression, Classification, Model Evaluation, Clustering, PyTorch
Data Visualization — Tableau, Matplotlib, Seaborn, Plotly
Databases & Tools — MongoDB, MySQL, PostgreSQL, Git, Docker
Web & APIs — HTML, CSS, JavaScript, MERN Stack (MongoDB, Express, React, Node.js), REST APIs
AI Tools — Large language models , Prompt Engineering , Context Engineering, LLM APIs , RAG

Languages

English — Fluent | **Hindi** — Fluent | **German** — B1